

Calibrate Once, Measure Within Support: Multicalibration for Unbiased Prevalence Estimation from Political Text

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Abstract

Political scientists increasingly use classifiers—and now large language models (LLMs)—to estimate how prevalent a category is across countries, time periods, and subgroups, such as the share of legislative attention on law and crime. We show that standard validation practice, reporting accuracy or AUC on a labeled sample, certifies neither valid prevalence estimates nor their cross-context comparability. When the target differs from the validation sample (covariate shift), an accurate classifier can still produce badly biased prevalence estimates, and the usual remedies—per-target reweighting, quantification, design-based correction—redo work for every new target. We show that *multicalibration*, calibration conditional on input features rather than on average, is sufficient for unbiased prevalence estimation under any covariate shift along the calibrated feature dimensions, yielding an instrument calibrated once and applied to any target within its support. A simulation and two applications—a cross-national LLM coding task on Comparative Agendas Project texts and a controlled employment benchmark—show near-zero bias under shift, where the default method (Classify & Count) is off by several points and label-shift quantifiers and misspecified importance weighting by up to roughly 20 points out of support. It works on the bare Yes/No labels researchers already collect, which match or beat calibrating on elicited probabilities.

Introduction

A growing share of empirical political science depends on machines reading text. Researchers use automated classifiers, and increasingly large language models (LLMs), to code policy agendas across countries (Baumgartner et al. 2006; Benoit et al. 2026), to detect protest and conflict events in news (Overos et al. 2024; King and Lowe 2003), to score democratic institutions (Weidmann et al. 2026), to categorize open-ended survey responses (Mellon et al. 2024), and to label social media at a scale that hand-coding cannot reach (Gilardi et al. 2023; Ornstein et al. 2025). In many of these studies the quantity of scientific interest is not the label on any single document but an *aggregate*: how prevalent a category is in a population. What fraction of a parliament’s questions concern law

and crime? How often does a topic appear in one country’s media versus another’s? These are *prevalence* (or *quantification*) problems (González et al. 2017; Hopkins and King 2010), and they are the workhorse output of text-as-data research.

The standard way researchers validate an automated coder is to report its discriminative performance—accuracy, F_1 , AUC, or agreement with human coders—on a labeled validation sample (Grimmer et al. 2022; Halterman and Keith 2025). A coder that clears a high bar is then applied to new corpora: other countries, earlier decades, different document types. This practice rests on an implicit assumption: that performance established on the validation sample transfers to the populations where the instrument is actually used. That assumption is rarely examined, and for prevalence estimation it is frequently wrong.

The problem is that discrimination and prevalence accuracy are different properties. A classifier can rank documents almost perfectly—separating positives from negatives—while systematically over- or under-stating *how many* positives there are. Worse, the direction and size of that prevalence error depend on the composition of the target population. When the target differs from the validation sample in its feature distribution (a *covariate shift*: the mix of languages, document types, eras, or authors changes, but the meaning of the category does not), a coder that produced unbiased prevalence estimates on the validation sample can produce badly biased ones elsewhere. Because covariate shift is the rule, not the exception, in comparative and over-time research—the very settings where a single instrument is applied to many populations—this is a first-order threat to measurement validity (Adcock and Collier 2001) that current validation practice does not detect.

Political methodology already offers tools that engage parts of this problem, but none delivers a *reusable* instrument. Quantification methods such as ReadMe (Hopkins and King 2010) and its successor (Jerzak et al. 2023) estimate category proportions directly, but under a *label-shift* assumption that is the wrong invariance for the covariate-shift regime we study. Design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023) debias a downstream estimate by combining cheap machine labels with a small set of gold labels *drawn from the target sample itself*—powerful and assumption-light, but target-specific by construction: each new population requires its own gold sample. Importance weighting handles covariate shift in principle but must re-estimate density ratios for every target and fails under positivity violations.

We show that *multicalibration* fills this gap. A device is multicalibrated when it is calibrated conditional on input features, not merely on average (Hébert-Johnson et al. 2018). The relevant statistical guarantee is the “universal adaptability” result of Kim et al. (2022): a multi-accurate predictor yields unbiased target means under any reweighting captured by its calibration features. We do not claim that guarantee as ours; our contribution is its translation into a measurement protocol for political text-as-data. That translation has two parts a methods reader should weigh separately. The conceptual part is a diagnostic taxonomy—covariate shift, not the label shift the ReadMe family assumes, is the regime of comparative and over-time text-as-data, which determines when each quantification method is valid. The empirical part, and the most practically consequential and non-obvious result, is that multicalibration on the *bare Yes/No labels* researchers already collect, plus observable metadata, matches or beats calibrating on elicited probability scores—so the correction comes from the calibration design, not from coaxing confidences out of the model. Concretely: calibrate once on a labeled source sample, then apply the instrument to any target whose features lie within the calibration support, with no target-specific re-estimation. The minimal condition is *multi-accuracy* (correct predictions on average within feature-defined subgroups); multicalibration is the stronger, practically attainable property that implies it and extends the

guarantee to score-conditional uses.

The remaining contributions are analytical and empirical. We make the analytical case explicit: every standard quantification and calibration method inherits a prevalence bias under covariate shift, and feature-conditional calibration removes it, with the trade-off stated precisely—a target-independent instrument bought with an ignorability/support assumption that design-based methods avoid. Two applications—a cross-national LLM coding task and a controlled benchmark with known ground truth—then show near-zero prevalence bias under shift, where the default Classify & Count is off by a few points and the methods that ignore feature-conditional structure (label-shift quantifiers, misspecified importance weighting) by up to roughly 20 percentage points out of support; within support, several standard methods are fine, so the failure is specifically a transportability one. The result is classifier-agnostic—LLMs are the leading instance, not the contribution.

Measurement, quantification, and shift in text-as-data research

Estimating aggregates from text. Extracting quantities from political text has a long lineage, from scaling policy positions with words as data (Laver et al. 2003) to automated event extraction matching human coders (King and Lowe 2003) to the supervised and unsupervised methods synthesized by Grimmer et al. (2022) and Grimmer and Stewart (2013). Throughout, the validation standard has been agreement with human coders: intercoder reliability for the human benchmark itself (Hayes and Krippendorff 2007), and accuracy or AUC for the machine. LLMs are the newest entrant, and a rapidly growing literature evaluates them as zero-shot coders of political concepts (Halterman and Keith 2025; Benoit et al. 2026; Ornstein et al. 2025; Weidmann et al. 2026), generally on the same discrimination metrics. Our argument is that for the prevalence estimands these studies target, agreement-on-average in a validation corpus is necessary but not sufficient: it does not certify that estimates transport to the populations of interest.

Quantification. The task of estimating class prevalence from imperfect classifiers is *quantification* in machine learning (González et al. 2017) and is the explicit target of ReadMe (Hopkins and King 2010) and ReadMe2 (Jerzak et al. 2023) in political science. These methods estimate category proportions without committing to per-document labels, and ReadMe2 improves robustness to changes in the document-feature distribution. Crucially, however, the ReadMe family and classical quantifiers (Adjusted Count, the Saerens-Latinne-Decaestecker EM algorithm (Saerens et al. 2002), distribution-matching methods (Maletzke et al. 2018)) assume *label shift*: the class-conditional feature distribution $P(X | Y)$ is stable while the class prior $P(Y)$ may change. That is the appropriate model when the label causes the features. In comparative text-as-data, the more natural model is the reverse—features (language, venue, era, content) drive the category, so $P(Y | X)$ is stable and $P(X)$ shifts. Under this covariate shift, label-shift quantifiers are not guaranteed to be unbiased, as our results confirm.

Using imperfect labels for valid inference. A complementary literature corrects downstream estimates computed from imperfect machine labels. Design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023) combine many cheap predictions with a small probability sample of gold labels to produce design-consistent estimates with valid uncertainty. These methods are assumption-light—they need no covariate-shift or ignorability assumption—but they are *target-specific*: the gold sample must come from the target population, so each new country, period, or subgroup requires fresh hand-coding. This is exactly the cost that comparative researchers, applying one instrument to dozens of corpora, are trying to avoid. We

treat these methods as complementary and return to the trade-off in the discussion. Related work on learned proxies in causal inference (Knox et al. 2022) and on the reliability of LLM annotations (Bisbee et al. 2024; Argyle et al. 2023; Rodriguez and Spirling 2022) shares our concern that imperfect measures can distort inference, but does not address the transportability of prevalence estimates across populations.

Calibration and its non-transfer. Calibration—that predicted probabilities match empirical frequencies—is the property that links a probabilistic coder to unbiased prevalence: if a device is globally calibrated on a population, the mean of its scores equals that population’s prevalence. But global calibration is a property of one population and need not survive a shift. Empirically, LLM calibration degrades two- to threefold under language shift even when accuracy is preserved (Yang et al. 2023) and varies widely across tasks (Ren et al. 2025). Confidence-elicitation methods—verbalized confidence (Tian et al. 2023), token log-probabilities (Kadavath et al. 2022), self-consistency (Wang et al. 2023)—and post-hoc recalibration target *global* calibration and so inherit this non-transfer. Multicalibration (Hébert-Johnson et al. 2018; Detommaso et al. 2024), by contrast, conditions on features and is therefore stable under reweighting of those features; this is the property we exploit.

Why calibration fails and multicalibration suffices

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a device $h(X) \in [0, 1]$. The estimand is the prevalence $\pi^* = P^*(Y = 1)$ in a target population, to be estimated from unlabeled target data $\{X_i\}$ and the device h . We assume *covariate shift*: $P(X)$ differs between calibration and target populations while $P(Y | X)$ is stable. This is the natural assumption when features causally precede the outcome, as in most measurement settings (the language and content of a document determine its topic, not vice versa). Label shift and concept drift are discussed under Limitations.

Global calibration is population-specific. A device is *globally calibrated* on a population if $\mathbb{E}[Y | h(X) = p] = p$ for all p . Under global calibration, $\mathbb{E}[h(X)] = \mathbb{E}[Y] = \pi$, so averaging predictions estimates prevalence without bias. But calibration on one population does not imply calibration on another. Partition the population into feature-defined groups G with weights w_G , and let $\epsilon_G = \mathbb{E}[h(X) - Y | X \in G]$ be the device’s mean signed error (predicted minus true) in group G . Global calibration requires only that these errors cancel *on average*: $\sum_G w_G \epsilon_G = 0$. Under covariate shift the weights become w_G^* , and the prevalence bias is

$$\hat{\pi}^* - \pi^* = \sum_G w_G^* \epsilon_G,$$

which is generally nonzero unless every $\epsilon_G = 0$. This identity is exact when the groups are fine enough that the density ratio is constant within each cell—the same condition that makes the universal-adaptability result below hold—and is otherwise a first-order approximation, since with coarse cells whose within-cell composition shifts, the source-estimated ϵ_G no longer equals the target’s within-cell error. A device can be perfectly calibrated on the validation sample—its group errors precisely offsetting under w_G —and badly biased on a target that reweights those same groups. High AUC offers no protection: AUC depends only on the ranking of scores, so it is invariant to all miscalibration, including the feature-conditional errors ϵ_G that drive prevalence bias under shift—a classifier can have near-perfect AUC and still be badly biased in prevalence.

Every standard method inherits this mechanism. Classify & Count, the Rogan-Gladen adjustment (Rogan and Gladen 1978), and PACC rely on error rates or conditional score means that are

weighted averages over groups and shift with composition. SLD/EMQ (Saerens et al. 2002) and distribution-matching quantifiers assume label shift and fail under covariate shift. Even global recalibration by isotonic regression removes only the average miscalibration, not the group-specific errors ϵ_G (Wu and Resnick 2024).

Feature-conditional calibration removes the bias. The fix is to drive every ϵ_G to zero, not merely their weighted sum. A predictor f is *multicalibrated* with respect to a class of groups $\mathcal{G}$ if $\mathbb{E}[Y \mid f(X) = v, X \in G] = v$ for all $G \in \mathcal{G}$ and values v (Hébert-Johnson et al. 2018). The condition strictly needed for unbiased prevalence on *every* target reweighting along $\mathcal{G}$ is weaker than full multicalibration—*multi-accuracy*, $\mathbb{E}[f(X) - Y \mid X \in G] = 0$ for every group $G \in \mathcal{G}$. (For one fixed target, only the weaker $\sum_G w_G^* \epsilon_G = 0$ is needed; multi-accuracy is what makes the instrument target-independent.) Multi-accuracy forces the group-averaged error to zero within each G ; it does not by itself imply pointwise accuracy $\mathbb{E}[f(X) \mid X = x] = \mathbb{E}[Y \mid X = x]$ unless $\mathcal{G}$ is rich enough to separate the relevant regions of X (in the limit, singletons). Its consequence for prevalence is precisely the Kim et al. (2022) “universal adaptability” guarantee: if f is multi-accurate with respect to $\mathcal{G}$, then for any target whose density ratio dP^*/dP is measurable with respect to $\mathcal{G}$ (equivalently, whose reweighting is captured by the calibration features),

$$\mathbb{E}^*[f(X)] = \mathbb{E}\left[\frac{dP^*}{dP}(X) f(X)\right] = \mathbb{E}\left[\frac{dP^*}{dP}(X) Y\right] = \mathbb{E}^*[Y] = \pi^*,$$

where the middle equality holds because the density ratio is constant within each $\mathcal{G}$ -cell, so the within-cell error $\epsilon_G = 0$ carries through. One instrument, calibrated once on source data, is thus unbiased on every target whose shift lies along the calibrated feature dimensions and within support, with no knowledge of the shift and no target-specific estimation.

Why target multicalibration rather than multi-accuracy. Multi-accuracy is the minimal sufficient condition, but multicalibration is the better practical target for three reasons. First, practical post-hoc algorithms produce multicalibrated predictors at no extra cost: MCGrad (Tax et al. 2026), which we use, fits the residual structure of X by gradient boosting and naturally approximates multicalibration. Second, multicalibration extends the guarantee to cases where multi-accuracy is insufficient: when the shift is mediated by the device’s own scores (e.g., score-based selection of documents to hand-code), when prevalence is estimated within score strata, or when scores enter a downstream regression. Third, a predictor multicalibrated over a rich group class $\mathcal{G}$ is automatically multi-accurate over any subgroup measurable with respect to $\mathcal{G}$, providing robustness to misspecifying which of the calibrated features drive the shift.

The scope condition. The guarantee rests on two distinct requirements that are worth separating. The first is *support/overlap*: the target’s feature values must lie within the calibration support, so that there is calibration data on which the predictor was corrected. The second, and more demanding, is that the calibration feature class $\mathcal{G}$ be *expressive enough that within-cell $P(Y \mid X)$ is stable across populations*—that the features capture every dimension along which the shift interacts with the outcome. The support condition is the analogue of the positivity that importance weighting needs; the expressiveness condition is an ignorability assumption discharged *once*, at calibration time, rather than re-estimated per target. Both are stronger than what design-based methods assume, and that is the price of target-independence. In text settings the expressiveness condition is the more fragile one: coarse metadata cells (the same “newspaper” document type in two eras with shifted writing style) may share all observed feature values while differing in latent content that drives Y , so that within-cell $P(Y \mid X)$ is not in fact stable. The natural mitigation is to enrich $\mathcal{G}$ with text-derived features—document embeddings (Rodriguez and Spirling 2022)—so that the

calibration cells track semantic content, not just metadata. When either condition fails—a genuinely novel feature value, or latent variation no feature captures—estimates degrade, as our out-of-support results show.

Simulation

We illustrate the mechanism in a setting simple enough to verify by hand. Data have a binary covariate $X \in \{0, 1\}$ and outcome Y with $P(Y = 1 \mid X = 1) = 0.85$ and $P(Y = 1 \mid X = 0) = 0.15$. A classifier produces deterministic, systematically biased scores—a 10% multiplicative bias within each stratum, so $\hat{p} = 0.135$ when $X = 0$ and $\hat{p} = 0.935$ when $X = 1$ —wrong within each stratum but corrected to the true prevalence by a single global recalibration step on the balanced training distribution $P(X = 0) = 0.5$. We estimate prevalence on target distributions with $P(X = 0)$ ranging from 0.01 to 0.99, holding all calibration parameters fixed at their training values, over 50 replications. In this two-valued example the score is a bijection of X , so score-conditional and feature-conditional calibration coincide; the contrast between them that drives the failure of score-only methods is the subject of the two applications, where scores coarsen the features.

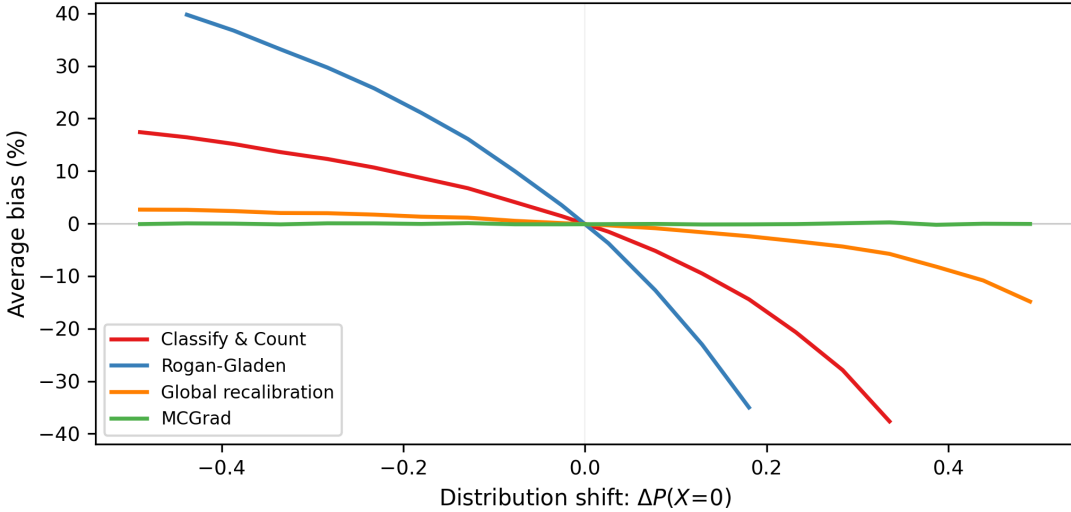


Figure 1. Relative prevalence bias under covariate shift (50 runs). The x-axis is the change in $P(X = 0)$ from the training value of 0.5. Classify & Count and Rogan-Gladen diverge with shift; global recalibration (multiplicative, the method of SI Section S1.6) shows moderate bias; the multicalibrated estimator stays near zero. Cropped at $\pm 40\%$; see SI Figure S3 for the full range and additional methods.

At the training distribution all methods are approximately unbiased (Figure 1). As the target shifts, they diverge: Rogan-Gladen is most unstable, its ratio form amplifying error beyond $\pm 40\%$; Classify & Count grows in the same direction; global recalibration shows moderate but real bias, up to roughly 15% at the most extreme shift (SI Figure S3). The multicalibrated estimator—here, stratum-specific additive corrections—maintains near-zero bias and the lowest RMSE across the entire range (SI Figures S2–S3), confirming the bias reduction does not come at the cost of variance.

Cross-national LLM coding under covariate shift

Our main application is a worked example of the modal text-as-data workflow: a single coding instrument applied across countries and languages. We use Claude Opus 4.6 to classify 30,000 political texts from the Comparative Agendas Project (Baumgartner et al. 2006) for whether they concern Law & Crime (CAP major topic 12). The data span six sub-populations across four countries and languages (5,000 documents each): Danish parliamentary questions, Spanish oral questions, U.S. congressional bills, Belgian newspaper articles, Spanish media articles, and Belgian TV news. The cross-national Law & Crime share is a substantive comparative-agendas estimand, and the design mirrors the real situation a researcher faces: code a concept once, then estimate its prevalence in many corpora. It is a realistic worked example on real comparative corpora—using the published CAP expert codes as the gold standard against which the LLM is scored—rather than a re-estimation of any single published study; its purpose is to show how the method behaves across the shift gradient practitioners actually encounter.

Design. We compare two LLM output modes spanning current practice. The first is the discrete Yes/No classification used in essentially all applied studies. The second is a probability score obtained by direct elicitation, where the model estimates $P(\text{Yes})$ without first committing to an answer—representing the confidence-elicitation literature’s recommendation. Discrimination is strong in both modes: AUC 0.960 (binary) and 0.987 (scores), with per-language AUCs of 0.983–0.994. The calibration set ($n \approx 13,400$) draws equally from four sub-populations, ensuring all four countries and languages are represented. Two sub-populations are held out as out-of-support targets—Spanish media and Belgian TV—which share country and language with the calibration data but introduce an unseen document type. We calibrate MCGrad on observable metadata only: country, document type, party, decade, and document length. For the binary condition, MCGrad receives the Yes/No label as a categorical feature with scores initialized at the calibration base rate; it never sees a probability from the model. For the score condition, it receives $P(\text{Yes})$ directly. We benchmark against Classify & Count, Rogan-Gladen, isotonic regression, importance weighting (IPW), and SLD (SI Table S2), plus the canonical political-science quantifier ReadMe (Hopkins and King 2010) (SI Section S7).

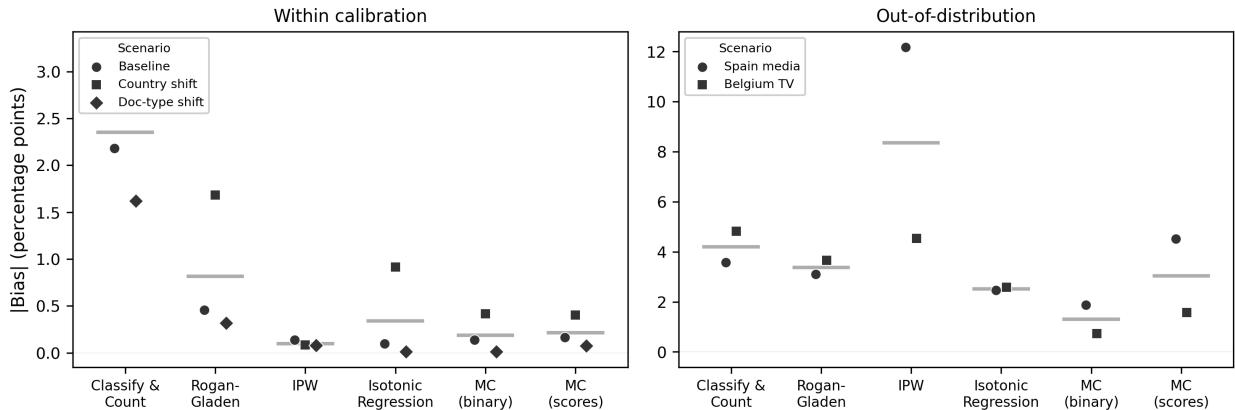


Figure 2. Absolute prevalence bias (percentage points) across the shift gradient: a no-shift baseline, two within-support shifts (country and document-type composition), and two out-of-support targets. Marker shape denotes scenario; horizontal lines are scenario means. The two multicalibration variants—MCGrad on bare Yes/No labels (MC binary) and on elicited probability scores (MC scores)—are adjacent for direct comparison; within support they are indistinguishable (both $\leq 0.4pp$,

a difference smaller than the roughly 0.4–0.6pp sampling error in the gold base rate), and the headline finding is that the binary variant requires no probability output at all. Full numbers in SI Table S2.

Results. The failure pattern is clear (Figure 2). Classify & Count—counting the model’s Yes labels, the default in applied work—is biased by +1.6 to +4.8pp across all scenarios. Rogan-Gladen reduces this within support but still misses by +3.1 to +3.7pp out of support. Isotonic regression is good within support (≤ 0.9 pp) but worse out of support (2.5–2.6pp). The two label-shift quantifiers we benchmark both fail under covariate shift. SLD is already biased at the no-shift baseline (+7.4pp, reflecting that the elicited scores are not the calibrated posteriors its EM step assumes) and the shift amplifies this to +21.6pp out of support. ReadMe (Hopkins and King 2010), the canonical political-science quantifier, is near-zero only without shift and badly biased and high-variance under it—by tens of percentage points (SI Section S7). ReadMe2 (Jerzak et al. 2023) refines ReadMe’s feature summary but estimates proportions under the same label-shift assumption, so it cannot escape this failure in principle (SI Section S7). Together these confirm that the label-shift family does not transport under covariate shift—now shown directly for the canonical quantifier, not inferred from SLD alone.

Multicalibration on the bare Yes/No labels achieves near-zero bias within the calibration distribution (≤ 0.4 pp) and degrades gracefully out of support: +0.7pp on Belgian TV and -1.9 pp on Spanish media. This is obtained from the Yes/No label and metadata alone—no probability scores. Multicalibration on probability scores matches it within support (≤ 0.4 pp) but is somewhat worse out of support (-4.5 pp on Spanish media, +1.6pp on Belgian TV). That bare labels match or beat elicited probabilities is our most practically important finding: it is the metadata features and the calibration sample, not the quality of the model’s confidences, that drive the correction. IPW achieves near-zero bias within support but fails severely out of support (-12.2 pp on Spanish media) because the target contains feature values absent from the source—a positivity violation—and, unlike multicalibration, it must re-estimate density ratios for each target.

Two caveats temper the within-support numbers. First, the gold standard is itself the CAP expert coding, so “true” prevalence is an estimate: the binomial sampling error on a sub-population base rate (5,000 documents) is about ± 0.4 – 0.6 pp (± 0.8 – 1.1 pp at 95%), so sub-1pp differences within support should be read as ties rather than rankings, and the substantive comparison is between near-zero-bias methods (multicalibration, IPW, isotonic within support) and the multi-point biases of Classify & Count and the label-shift quantifier. We report this binomial term rather than the full-pipeline bootstrap of step 4 below because here it dominates: with a 13,400-document calibration set the calibration-fitting variance is small relative to the gold-standard sampling error, and the ranking of methods by point bias is robust to it; the ACS benchmark, where truth is exact, exercises the full bootstrap, whose 200-iteration RMSE closely tracks the absolute bias in every scenario (Table S1). Second, the field setting lacks an oracle by construction; the controlled benchmark below supplies the identification-clean confirmation where truth is known exactly. We replicate the entire analysis with an open-weight model, Llama 3.3 70B, in SI Section S2: the pattern holds (MCGrad ≤ 0.2 pp within calibration), confirming the findings are not specific to a proprietary model and are reproducible without API access.

Controlled validation with known ground truth

The CAP application is the realism demonstration; this controlled benchmark is the identification-clean confirmation, where ground truth is known by construction and the base classifier is a

conventional, well-behaved model rather than an LLM. It is the only setting with an exact oracle, so it isolates the method’s behavior from any noise in a gold standard. Using the American Community Survey via the *folktables* package, we predict employment from 16 sociodemographic features with logistic regression, training on eight states (2016–2018; approximately 1.5M observations) and calibrating on a held-out set ($n \approx 644,000$). Because employment rates vary sharply by age (76% for ages 25–54 versus 17% for 65+), we construct covariate shifts by resampling the test set to be young-skewed, old-skewed, or bimodal, yielding true employment rates from 12.8% to 46.0%; all calibration parameters are fixed across scenarios. We evaluate both on in-distribution states and on six held-out states.

The pattern matches the simulation and CAP results (SI Table S1, visualized in SI Figure S1). Rogan-Gladen fails by 12–19pp and SLD comparably; Classify & Count and isotonic regression show moderate but growing bias (up to 8pp); IPW is good on simple shifts (≤ 1.2 pp) but fails on the bimodal shift (+4.7pp in-distribution, +6.3pp out-of-distribution) where the density ratio is hard to model. Multicalibration achieves ≤ 0.27 pp bias across all in-distribution scenarios, including the bimodal shift, and degrades only modestly out of distribution (0.88–1.35pp), reflecting geographic shift along a dimension the calibration set did not span. Across both applications the story is consistent: multicalibration is near-unbiased when the target’s features lie within the calibration support and degrades predictably when they do not—the scope condition has visible, interpretable bite rather than silent failure.

What this means for practice

The result yields a concrete recipe that fits the workflow researchers already use.

1. **Code as usual.** Prompt the LLM (or run any classifier) for a Yes/No label per document. No probability scores, confidence elicitation, or model internals are required.
2. **Collect a calibration sample that spans the shift dimensions.** Hand-code a modest labeled sample that *covers* the feature dimensions along which your targets will differ—country, language, document type, era. Coverage of these dimensions, not raw size, is what the guarantee requires (13,400 documents sufficed for CAP).
3. **Multicalibrate on observable metadata.** Fit a multicalibration algorithm conditioning on the document metadata (and the model’s label or score, if available). This needs no access to the model’s weights or logits.
4. **Estimate prevalence by averaging calibrated scores** on any target whose features lie within the calibration support, with no re-fitting. For uncertainty, bootstrap the full pipeline—resample the calibration set, refit the calibrator, and recompute the target average—to obtain a standard error and confidence interval; this is the procedure we use for the ACS RMSEs (200 iterations) and it propagates both calibration-fitting and target-sampling error into the reported estimate.

Which tool when. The choice among methods is a choice about where labeled data comes from. If you can draw a gold-label probability sample *from each target* you analyze, design-based supervised learning (Egami et al. 2023) or prediction-powered inference (Angelopoulos et al. 2023) are efficient and require no covariate-shift assumption—prefer them. If instead you must measure across *many* targets without per-target hand-coding—the comparative-agendas situation, where re-coding thousands of documents per new country is infeasible—multicalibration provides a single

reusable instrument, at the cost of the ignorability/support assumption. ReadMe-family quantifiers (Hopkins and King 2010; Jerzak et al. 2023) are the right choice under label shift (the prior changes, the class-conditional language does not), but fail under covariate shift, as we show directly (SI Section S7); multicalibration is the right choice under covariate shift.

Diagnostics. Because the guarantee is support-limited, verify coverage explicitly rather than reporting AUC alone. A concrete check: cross-tabulate the target’s feature cells (country $\times$ document type $\times$ era, plus binned length) against the calibration cells, and flag any target cell that is empty or near-empty in calibration; those cells are extrapolation and their contribution to the target average is unguaranteed. The amount of calibration data needed is whatever makes these cells adequately populated—coverage of the shift dimensions, not raw size, is the binding constraint (13,400 documents sufficed for CAP because the four calibration sub-populations spanned the relevant country and language cells). The out-of-support degradation in both applications is the cautionary case: a feature dimension—an unseen document type—that the calibration set did not cover, which the cross-tabulation would have flagged in advance. The practical implication is a design rule: build calibration data to span the dimensions along which your populations vary.

Discussion

Estimating how prevalent a category is across populations is among the most common uses of text-as-data, and it is more fragile than current practice acknowledges. Discriminative validation certifies that a coder separates classes; it is silent about whether prevalence estimates transport. We have shown that the property that makes them transport is calibration conditional on features—multi-accuracy as the minimal sufficient condition, multicalibration as its practical realization—and that this property turns a validated coder into a measurement instrument that can be calibrated once and applied to many populations. The result connects two literatures that developed apart: quantification, which corrected aggregate error rates but never engaged feature-conditional calibration, and multicalibration, which studied individual-level prediction and fairness but not population-level inference.

The central practical payoff is target-independence. Importance weighting, design-based supervised learning, and prediction-powered inference all redo work for each target—density ratios or gold samples drawn anew per population. A multicalibrated instrument is calibrated once and deployed without target-specific estimation, the property comparative and over-time research most needs. That advantage is bought with an assumption: the calibration features must capture the shift, and targets must lie in support. We view stating this scope condition—and showing its bite through honest out-of-support results—as a strength rather than a caveat to bury.

Several limitations follow. First, the guarantee holds only along calibrated dimensions; novel feature values degrade estimates, so calibration data must be designed to span anticipated variation. Second, multicalibration requires labeled calibration data; in zero-shot settings these labels reintroduce the hand-coding that LLMs were meant to avoid, though far less of it and only once. When per-target gold labels are available, design-based methods are the better tool and the two approaches compose. Third, we assume covariate shift; under concept drift ($P(Y | X)$ itself changes across contexts) no purely statistical correction substitutes for new labels. Finally, our applications use settings with measurable ground truth; practitioners often lack it, which makes the support diagnostics above all the more important. The accuracy and AUC that text-as-data research reports remain necessary—but for valid prevalence estimates across contexts, they are not sufficient.

Methods

Simulation. We generate $n = 10,000$ observations per run with $X \in \{0, 1\}$, $Y \sim \text{Bernoulli}(P(Y | X))$, $P(Y = 1 | X = 1) = 0.85$, $P(Y = 1 | X = 0) = 0.15$; the classifier returns $\hat{p}(X=0) = 0.135$, $\hat{p}(X=1) = 0.935$ (10% multiplicative bias per stratum). For each of 50 runs we draw fresh calibration data at $P(X = 0) = 0.5$, estimate all parameters, and evaluate bias and RMSE on 20 targets with $P(X = 0) \in [0.01, 0.99]$. Full definitions of all seven methods are in SI Section S1.

Comparative Agendas Project. We sample 30,000 documents (5,000 from each of six sub-populations) coded by Claude Opus 4.6 in two independent campaigns—binary Yes/No and direct probability elicitation—run separately to avoid anchoring. The calibration set ($n \approx 13,400$) draws equally from Denmark questions, Spain questions, U.S. bills, and Belgium newspaper; Spanish media and Belgian TV are held out as out-of-support targets sharing country and language but differing in document type. MCGrad conditions on country, document type, party, decade, and document length. For the binary condition the Yes/No label is a categorical feature with base-rate initialization; for the score condition $P(\text{Yes})$ is the input. Benchmarks: Classify & Count (fraction of Yes), Rogan-Gladen (TPR/FPR from calibration), isotonic regression on scores, IPW (density ratios via logistic regression on country, document type, decade, length), SLD (SI Table S2), and ReadMe (Hopkins and King 2010)—a classifier-free quantifier estimated from binary word-presence features on the document text over random feature subsets, run both in a from-scratch implementation and via the official IQSS `readme` package (SI Section S7; `cap_analysis/readme_baseline.py`). The Llama 3.3 70B replication is in SI Section S2.

American Community Survey. Using *folktables*, we predict employment with standardized logistic regression on 16 features; train on TX, MI, PA, OH, IL, GA, NC, VA (2016–2018); calibrate on a held-out set ($n \approx 644,000$); test in-distribution ($n \approx 920,000$) and on held-out states CA, NY, FL, WA, AZ, CO. Age-shifted targets are produced by importance-weighted resampling; RMSE via 200-iteration bootstrap. Post-hoc calibration uses isotonic regression and MCGrad with categorical and numerical features.

Multicalibration algorithm. Our claims are algorithm-agnostic: any procedure that produces a multi-accurate (a fortiori multicalibrated) predictor over the feature class delivers the same guarantee, and several exist (Hébert-Johnson et al. 2018; Gopalan et al. 2022; Detommaso et al. 2024). We use MCGrad (Tax et al. 2026) for its convenience on tabular metadata; it operates in logit space, iteratively fitting gradient-boosted trees on label residuals with the current prediction included as an input feature, so splits discover miscalibrated regions in the joint space of features and score levels, with early stopping to prevent overfitting. Full algorithmic detail is in SI Section S1.7. An open-source implementation is at <https://mcgrad.dev>.

Data availability. Replication code is available at https://github.com/facebookresearch/multicalibrated_llm_measurement and will be deposited in the Political Analysis Dataverse on acceptance. ACS data is public via *folktables*; CAP data is public at <https://www.comparativeagendas.net>. Claude Opus inference outputs for all 30,000 main-text documents and Llama 3.3 70B outputs for all 105,000 documents used in SI Section S2 are included in the replication materials. A disclosure of generative-AI use in preparing this manuscript is provided in SI Section S8.

Competing interests. The authors declare no financial competing interests. In the interest of full disclosure, several authors are affiliated with Meta Platforms, and a subset are authors of the MCGrad algorithm used here; the open-source implementation is maintained by their team. The conceptual claim and the empirical guarantee are algorithm-agnostic, and any multi-accuracy or

multicalibration fitter can be substituted for MCGrad. The open-weight Llama 3.3 70B replication (SI Section S2) shows the method does not depend on proprietary model access.

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